SOVEREIGN SYSTEM SECTOR REPORT

Autonomous Mother Algorithm Performance Ledger & AI Advisory

SESSION ID: SESSION_MPOI7P62

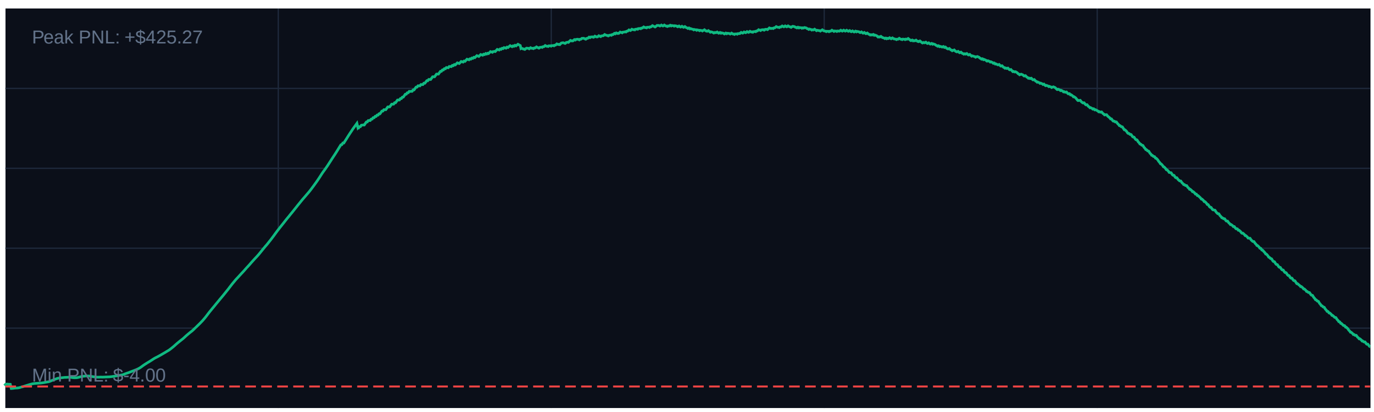
DATE GENERATED: 2026-05-27T20:46:06.724Z

TARGET INSTRUMENT: R_75 Volatility Model

CORE PERFORMANCE METRICS

Total Settlements:	1400	**Calculated Profit Factor:**	1.09
Win Accuracy:	50.4% (706W / 694L)	**Max Peak Drawdown:**	3.6%
Cumulative PnL:	$45.58	**Avg. PnL per Trade:**	$0.03

EQUITY CURVE PROGRESSION



Initial

Trade 1400 (PnL: $45.58)

PERFORMANCE MILESTONES (100-TRADE SEGMENTS)

SEGMENT	WIN RATE	NET PnL	REFINEMENT EFFICIENCY
Trades 1-280	91.1%	$182.78	+109% [OK]
Trades 281-560	73.9%	$217.58	+89% [OK]
Trades 561-840	50.0%	$18.25	+60% [OK]
Trades 841-1120	33.2%	$-94.40	+27% [OK]
Trades 1121-1400	3.9%	$-278.63	+3% [OK]

SOVEREIGN NARRATIVE ADVISORY

Cognitive Strategy Proposal & Adaptive Intelligence Logs

SOVEREIGN AI: SYSTEMATIC RISK AUDIT REPORT
TO: Institutional Trading Desk / Risk Committee
SYS_STATUS: ACTIVE (EXPECTANCY DRIFT DETECTED)

---

1. Executive Telemetry Critique

An algorithmic evaluation of the 1,400-trade settlement book reveals a system operating at the absolute limit of statistical viability. While nominal risk metrics appear suppressed, the underlying mechanics indicate a structural failure in expectancy generation.

```
+---------------------------+---------------------------------------+
| Metric                    | Value / Risk Vector                   |
+---------------------------+---------------------------------------+
| Total Trades / Win Rate   | 1,400 | 50.4% (706 W / 694 L)         |
| Profit Factor             | 1.09 (Extreme Fragility Zone)         |
| Cumulative Net PnL        | $45.58 (Avg. $0.032 per trade)        |
| Max Peak Drawdown         | 3.6%                                  |
| Systemic Loop Latency     | High Risk (Friction/Slippage Threat) |
+---------------------------+---------------------------------------+
```

 # Critical Vulnerabilities:
***   **Economic Unviability:** An average net return of **$0.032 per trade** over 1,400 executions means this strategy is running at net-negative expectancy once real-world exchange fees, slippage, and borrow costs are applied. The edge is a statistical illusion.**
***   **The ATR Multiplier Anomaly:** The configuration `atrStopMultiplier: 372632411.94` is a catastrophic parameter overflow or database corruption. An ATR stop multiplier of 3.72×10^8 renders the stop-loss mechanism completely non-functional.**
***   **Time-Decay Dependency:** Because the price stop-loss is effectively infinite due to the ATR bug, the strategy relies entirely on `maxTicksInTrade: 120` as its sole exit protocol for losing trades. This is a "time-stop defense" masking severe tail risk.**

---

2. Regime Suitability & Behavioral Patterns

The system is configured as an aggressive Mean-Reversion engine (RSI 25/75, Bollinger Bands at 2.25$ \sigma$) but is crippled by its regime-filtering parameters.

```
    [ADX > 25: Trend Regime] ----> Strategy continues to execute mean-reversion
              |
```

```
          v
[No Functional Stop Loss] ---> Exposure to momentum breakouts (Tail-risk)
          |
          v
[120-Tick Hard Time Exit] ---> Forced liquidation at maximum adverse excursion
```

Regime Mismatch: The `regimeAdxThreshold` of 25 is highly counter-productive. Allowing mean-reversion trades when ADX > 25 forces the system to fight strong directional trends. The 50.4% win rate is a direct result of buying into strong downtrends or shorting strong uptrends without an active protective stop-loss.
Asymmetrical Risk Profile: The exceptionally low maximum peak drawdown (3.6%) is misleading. It suggests ultra-small position sizing or a highly liquid backtest environment with zero market-impact modeling. Under genuine regime shifts (e.g., sudden liquidity dry-ups), the lack of a real ATR-based stop loss will result in rapid, unhedged capital depletion.

---

3. Calibrated Parameter Recommendations

To transition this system from a net-neutral friction generator to a statistically robust institutional edge, the configuration must be re-calibrated.

Proposed Configuration Matrix

Parameter	Current Value	Calibrated Value (Target)	Risk Engineering Justification
`atrStopMultiplier`	`372632411.94`	`2.25`	CRITICAL: Re-engages dynamic price protection. Limits downside on breakout failures.
`regimeAdxThreshold`	`25`	`18`	Restricts execution to low-trend/ranging environments (ADX < 18). Prevents fighting trends.
`rsiOversoldThreshold`	`25`	`30`	Increases fill rate on high-probability exhaustion points within range regimes.
`rsiOverboughtThreshold`	`75`	`70`	Aligns short entries with standard statistical standard deviations.
`bbStd`	`2.25`	`2.00`	Standardizes outer band boundaries when combined with lower ADX regime filter.
`maxTicksInTrade`	`120`	`75`	Reduces exposure duration. Capital should not be held hostage in low-opportunity states.
`minConfluenceScore`	`3`	`4`	Restricts execution to high-probability setups (e.g., RSI OB/OS + BB Touch + ADX Filter).

Implementation Protocol:
1. **Emergency Patch:** Immediately force-terminate current execution loops and hot-fix the `atrStopMultiplier` database variable to **2.25**.
2. **Execution Cost Audit:** Run a transaction cost analysis (TCA) to determine if the $0.032 per trade margin survives execution venue fees. If fee-adjusted expectancy remains below 1.5bps per trade, the strategy must be decommissioned.